



Practice Problem Set -07 (Linear Algebra)

Are the following vectors linearly dependent? If so, find the relation between them

1. $(2, 1, 1), (2, 0, -1), (4, 2, 1)$
2. $(1, -1, 0), (0, 1, -1), (0, 0, 1)$
3. $(1, 1, -1, 1), (1, -1, 2, -1), (3, 1, 0, 1)$
4. $(1, 2, 4), (2, -1, 3), (0, 1, 2), (-3, 7, 2)$
5. $(6, 0, -1, 3), (2, 2, 5, 0), (-4, -4, -4, -4)$
6. $(9, 8, 7, 6, 5), (9, 7, 5, 3, 1)$
7. $(1, 2, 3, 4), (2, 3, 4, 5), (3, 4, 5, 6), (4, 5, 6, 7)$
8. $(2, 1), (3, 2), (1, 1), (2, 3)$
9. $(2, 0, 0, 7), (2, 0, 0, 8), (2, 0, 0, 9), (2, 0, 1, 0)$
10. $(2, -1, 4), (0, 1, 2), (6, 1, 14), (4, 0, 12)$
11. $(1, 2, 1), (2, 1, 4), (4, 5, 6), (1, 8, 3)$
12. $(3, 2, 4), (1, 0, 2), (1, -1, 0)$
13. Express the vector $(3, 7, -4) \in R^3$ as a linear combination of $x_1 = (1, 2, 3)$, $x_2 = (2, 3, 2)$, $x_3 = (3, 5, 6)$.
14. Express the vector $(2, 3, -1, -1) \in R^4$ as a linear combination of $x_1 = (3, 1, 3, 2)$, $x_2 = (1, -1, -2, -4)$, $x_3 = (6, 3, 0, -7)$.
15. let $T: R^3 \rightarrow R^2$ be the transformation $T(x, y, z) = (x, y)$, Show that T is linear.
16. let $T: R^3 \rightarrow R^3$ be the transformation $T(x, y, z) = (x - y, z, y - x)$, Show that T is linear.
17. let $T: R^2 \rightarrow R^3$ be the transformation $T(x, y) = (x, x + y, y)$, Show that T is linear.
18. let $T: R^2 \rightarrow R^3$ be the transformation $T(x, y) = (x + y, x + 2y, y)$, Show that T is linear.
19. let $T: R^3 \rightarrow R^3$ be the transformation $T(x, y, z) = (x + 1, z + 2, y)$, check whether T is linear.
20. let $T: R^2 \rightarrow R^2$ be the transformation $T(x, y) = (0, x)$, check whether T is linear..